

NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Mid-Term-Spring-2026

Level: Master

Programme: ME/MSc.

Course: Database Engineering

Year: 2026

Full Marks :60

Time : 2 hrs

Candidates are required to give answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions

1.a) Design an appropriate Entity-Relationship (ER) model for Pokhara University which wants to design a database for its Examination Section to manage students, courses, exams, and results. Then, convert the ER model into a relational schema with necessary DDL SQL Queries and Keys. 7+8

13 b) Consider the following relations:

Student(SID, Name, DeptID)

Department(DeptID, DeptName)

Course(CID, Title, DeptID)

Enrollment(SID, CID, Marks)

Write both SQL queries and Relational Algebra for the following:

- i. Display names of students who scored more than 80 marks.
- ii. List student names along with course titles they are enrolled in.
- iii. Find the average marks of each course.
- iv. Display department names with the number of students in each department.

2 a) Create a materialized view to calculate monthly total sales made by customers. Also explain different caching strategies for materialized views. 7+8

- 2) b) Compare and contrast B-tree indexing, Hash indexing, and Bitmap indexing. Explain, with one example, how Bitmap indexing is often considered suitable for low-cardinality attributes.
- 3) 3.a) Explain the PACELC trade-off and how it extends the CAP Theorem. Illustrate your answer by analyzing the design choices and trade-offs in an e-commerce platform. 8+7
- 1) b) Explain the working of distributed transaction protocols with reference to Two-Phase Commit (2PC) and Three-Phase Commit (3PC). Compare them with consensus algorithms such as Paxos or Raft, and analyze their suitability for ensuring consistency and fault tolerance in distributed database systems.
- 4.a) A social network stores users and their friendships in a graph database. Write a query (using a suitable graph query language) to find all friends-of-friends of a given user. Explain how graph traversal works in this scenario and discuss the advantages of using a graph database for such operations. 8+7
- 2) b) How NoSQL differs from Relational Database? Explain different types of NoSQL Data Models. Also, explain with an example, how Aggregation is performed in NoSQL.